

# Identities, Multiple Angles, and Exact Solutions

## Answer Key

### Precalculus

#### Quick Answers

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|------------------------------------|------------------------------------------------------|
| 1. $\sin(x)$                       | 7. $0, \frac{2\pi}{3}, \frac{4\pi}{3}$               |
| 2. $\cos(x)$                       | 8. $\tan(x)$                                         |
| 3. $1 - 2\sin^2(x)$                | 9. $2\sec^2(x)$                                      |
| 4. $\frac{\pi}{6}, \frac{5\pi}{6}$ | 10. $0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}$          |
| 5. $\frac{24}{25}$                 | 11. $-\frac{119}{169}$                               |
| 6. $2\sin(x)$                      | 12. $\frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$ |
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#### Curriculum

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**Level:** Precalculus

HSF-TF.A.2, HSF-TF.B.5 – *problem 4*

HSF-TF.C.8 – *problems 1, 2, 3, 5, 6, 7, 8, 9, 10, 11, 12*

*Difficulty 1–5 of 5 across 12 tagged checks.*

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#### Common wrong answers

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5. *If they got 1.2: doubled the sine instead of using the double-angle formula*

11. *If they got -0.77: doubled the cosine instead of using the double-angle formula*

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**1. Simplify**  $\frac{1 - \cos^2 x}{\sin x}$ .

**Step 1.** The numerator is one of the two rearrangements of the Pythagorean identity, so replace it rather than expanding anything.

**Step 2.**  $1 - \cos^2 x = \sin^2 x$ , so the expression is  $\frac{\sin^2 x}{\sin x}$ .

**Step 3.** Cancel one factor of  $\sin x$ :  $\boxed{\sin x}$  (valid where  $\sin x \neq 0$ ).

**2. Simplify**  $\sec x - \sec x \sin^2 x$ .

**Step 1.** Both terms share  $\sec x$ ; factoring first exposes a Pythagorean form instead of leaving two unrelated terms.

**Step 2.**  $\sec x(1 - \sin^2 x) = \sec x \cos^2 x$ .

**Step 3.**  $\sec x = \frac{1}{\cos x}$ , so  $\frac{\cos^2 x}{\cos x} = \boxed{\cos x}$ .

**3. Rewrite  $\cos 2x$  in terms of  $\sin x$  alone.**

**Step 1.** There are three forms of the cosine double angle. Choosing is the whole point: pick the one whose surviving function is the one you were asked for.

**Step 2.** From  $\cos 2x = \cos^2 x - \sin^2 x$ , substitute  $\cos^2 x = 1 - \sin^2 x$ :  $(1 - \sin^2 x) - \sin^2 x$ .

**Step 3.**  $\cos 2x = \boxed{1 - 2\sin^2 x}$

**4. Solve  $2\sin x - 1 = 0$  on  $[0, 2\pi)$ .**

**Step 1.** One trigonometric function, first power — isolate it and read the unit circle; no identity is needed here.

**Step 2.**  $\sin x = \frac{1}{2}$ .

**Step 3.** Sine is positive in quadrants I and II. The reference angle is  $\frac{\pi}{6}$ , so the quadrant II partner is  $\pi - \frac{\pi}{6} = \frac{5\pi}{6}$ .

**Step 4.**  $x = \frac{\pi}{6}, \frac{5\pi}{6}$

**5.  $\sin \theta = \frac{3}{5}$  in quadrant I; find  $\sin 2\theta$ .**

**Step 1.** A double angle with both  $\sin \theta$  and  $\cos \theta$  known calls for  $\sin 2\theta = 2\sin \theta \cos \theta$  — no quadrant work is left, since quadrant I makes both values positive.

**Step 2.**  $2 \left( \frac{3}{5} \right) \left( \frac{4}{5} \right) = \frac{24}{25}$ .

**Step 3.**  $\sin 2\theta = \boxed{\frac{24}{25}}$

*If they got 1.2: doubled the sine itself.  $\sin 2\theta$  is never  $2\sin \theta$  — that value even exceeds the range of sine.*

**6. Simplify  $\tan x \cos x + \sin x$ .**

**Step 1.** A lone tangent among sines and cosines is the signal to rewrite it as a quotient; nothing else in the expression needs touching.

**Step 2.**  $\tan x \cos x = \frac{\sin x}{\cos x} \cos x = \sin x$ .

**Step 3.** The expression is  $\sin x + \sin x = \boxed{2\sin x}$ .

**7. Solve  $2\cos^2 x - \cos x - 1 = 0$  on  $[0, 2\pi)$ .**

**Step 1.** One function appears, but squared — so this is a quadratic in  $\cos x$ . Factor rather than isolating.

**Step 2.**  $(2\cos x + 1)(\cos x - 1) = 0$ , giving  $\cos x = -\frac{1}{2}$  or  $\cos x = 1$ .

**Step 3.**  $\cos x = -\frac{1}{2}$  has reference angle  $\frac{\pi}{3}$  and is negative in quadrants II and III, giving  $\frac{2\pi}{3}$  and  $\frac{4\pi}{3}$ .  $\cos x = 1$  gives  $x = 0$ .

**Step 4.**  $x = 0, \frac{2\pi}{3}, \frac{4\pi}{3}$

**8. Simplify**  $\frac{\sin 2x}{1 + \cos 2x}$ .

**Step 1.** Both double angles must go, and the denominator decides which cosine form to use:  $\cos 2x = 2\cos^2 x - 1$  is the one that makes  $1 + \cos 2x$  collapse.

**Step 2.** Numerator:  $\sin 2x = 2\sin x \cos x$ . Denominator:  $1 + (2\cos^2 x - 1) = 2\cos^2 x$ .

**Step 3.**  $\frac{2\sin x \cos x}{2\cos^2 x} = \frac{\sin x}{\cos x} = \boxed{\tan x}$

**9. Combine**  $\frac{1}{1 - \sin x} + \frac{1}{1 + \sin x}$ .

**Step 1.** The two denominators are conjugates, so their product is a difference of squares — which is exactly the shape the Pythagorean identity simplifies.

**Step 2.** Common denominator  $(1 - \sin x)(1 + \sin x) = 1 - \sin^2 x = \cos^2 x$ ; the numerators add to  $(1 + \sin x) + (1 - \sin x) = 2$ .

**Step 3.**  $\frac{2}{\cos^2 x} = \boxed{2\sec^2 x}$

**10. Solve**  $\sin 2x = \sin x$  **on**  $[0, 2\pi)$ .

**Step 1.** The two sides carry different angles, so expand the double angle first. Then move everything to one side — dividing by  $\sin x$  would destroy the solutions where  $\sin x = 0$ .

**Step 2.**  $2\sin x \cos x - \sin x = 0$ , so  $\sin x(2\cos x - 1) = 0$ .

**Step 3.**  $\sin x = 0$  gives  $x = 0$  and  $x = \pi$ .  $\cos x = \frac{1}{2}$  gives  $x = \frac{\pi}{3}$  and  $x = \frac{5\pi}{3}$ .

**Step 4.**  $\boxed{x = 0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}}$

**11.**  $\cos \theta = -\frac{5}{13}$  **in quadrant II; find**  $\cos 2\theta$ .

**Step 1.** Only the cosine is known, so choose the double-angle form built from cosine alone:  $\cos 2\theta = 2\cos^2 \theta - 1$ . The quadrant does not matter here, because the cosine is squared.

**Step 2.**  $2\left(-\frac{5}{13}\right)^2 - 1 = 2 \cdot \frac{25}{169} - 1 = \frac{50}{169} - \frac{169}{169}$ .

**Step 3.**  $\cos 2\theta = \boxed{\frac{-119}{169}}$

*If they got  $-0.77$ : doubled the cosine instead of squaring it first.*

**12. Challenge: solve**  $2\cos^2 x + \sin x = 1$  **on**  $[0, 2\pi)$ .

**Step 1.** Two different functions appear and one is squared, so the first move is an identity that leaves a single function — replace  $\cos^2 x$  with  $1 - \sin^2 x$  so everything is in sine.

**Step 2.**  $2(1 - \sin^2 x) + \sin x = 1$ , so  $2 - 2\sin^2 x + \sin x - 1 = 0$ , i.e.  $2\sin^2 x - \sin x - 1 = 0$ .

**Step 3.** Factor:  $(2\sin x + 1)(\sin x - 1) = 0$ , so  $\sin x = -\frac{1}{2}$  or  $\sin x = 1$ .

**Step 4.**  $\sin x = 1$  gives  $x = \frac{\pi}{2}$ .  $\sin x = -\frac{1}{2}$  is negative in quadrants III and IV, with reference angle  $\frac{\pi}{6}$ , giving  $\frac{7\pi}{6}$  and  $\frac{11\pi}{6}$ .

**Step 5.**  $\boxed{x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}}$